DATA 201 / MODEL PITCH

Can a bank predict who will subscribe?

A practical classification problem: focus outreach without missing interested customers.

45,211
Customer records

11.7%
Subscribed

4
Models compared

The trade-off Avoid wasted calls • Find customers who might say yes

Most customers did not subscribe

45,211 bank marketing records • 17 original columns

Outcome distribution

88.3% **11.7%**



No · 39,922 Yes · 5,289

What the model uses

- Customer information
- Financial characteristics
- Campaign and contact history

Why it matters: Predicting “No” for everyone already yields 88.3% accuracy.

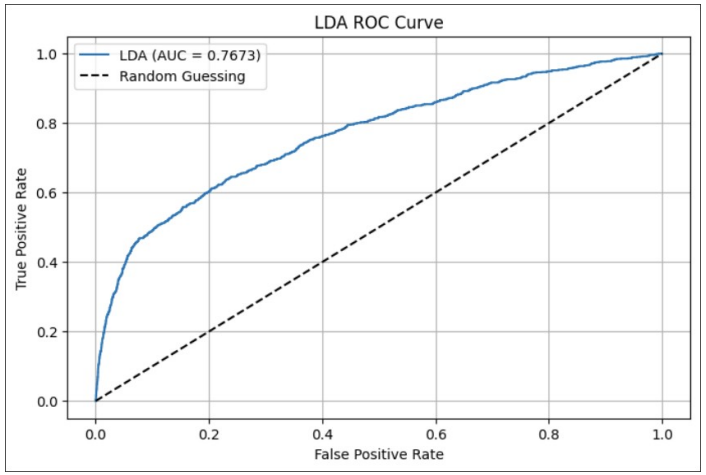
Accuracy hides important differences

Test-set results at the default classification threshold (0.50)

MODEL	ACCURACY	PRECISION	RECALL	F1
Logistic regression	89.33%	66.55%	17.67%	27.93%
LDA	**89.19%**	**57.49%**	**29.02%**	**38.57%**
QDA	88.26%	49.70%	31.10%	38.26%
KNN (K = 31)	89.02%	70.19%	10.68%	18.54%

LDA: highest F1 score

LDA ROC curve



ROC-AUC 0.7673

LDA has the best F1, balancing false alarms and missed subscribers.

Pilot LDA — then test the business impact

Best F1 score among the four evaluated classifiers

38.57%

LDA F1 score

307	227
Actual subscribers found	False positives

Why LDA?

TRADE-OFF	Balances unnecessary contacts and missed subscriptions.
LIMITATION	Class imbalance: many subscribers are still missed.
NEXT STEP	Measure contact costs, subscription value, and adjust the decision threshold.

Decision: start with LDA, then validate whether the added subscriptions justify the marketing cost.